



Signals and Systems Project - Phase 2

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**Time-Resolved Analysis of Phase-Amplitude Coupling (PAC) in EEG
During Olfactory Stimulation**

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1 Phase 2: Phase-Amplitude Coupling (PAC) Analysis

1.1 Building on Phase 1: Toward PAC

In the first phase of the project, the EEG data recorded during an olfactory stimulation task were preprocessed to remove artifacts, extract events, and prepare the signals for analysis. This included filtering, re-referencing, ICA-based artifact rejection, epoching, and trial-level cleaning. The preprocessed data were then analyzed to investigate power dynamics over time using time-frequency decomposition (Short-Time Fourier Transform), focusing particularly on the theta and gamma bands. These bands were selected based on their relevance to sensory processing and memory-related neural activity.

That initial power analysis provided insight into the activation levels of specific brain rhythms under different conditions and subject groups (Healthy Control, MCI, Mild Alzheimer's Disease). However, power alone does not reveal how these rhythms interact. The second and final phase now focuses on **Phase-Amplitude Coupling (PAC)** to explore how the amplitude of high-frequency oscillations (e.g., gamma) may be modulated by the phase of slower rhythms (e.g., theta). PAC reflects cross-frequency communication in the brain and is a candidate biomarker for cognitive decline. This phase introduces two methods to quantify PAC: **Mean Vector Length (MVL)** as the required method, and **Modulation Index (MI)**.

1.2 What is Phase-Amplitude Coupling?

Phase-Amplitude Coupling (PAC) is a form of cross-frequency coupling in which the amplitude envelope of a high-frequency brain oscillation is systematically modulated by the phase of a slower oscillation. In functional terms, it reflects how integrative processes (associated with low frequencies like theta) structure the timing of local, fast computations (associated with high-frequency gamma activity). For example, in healthy cognition, gamma bursts related to perception or memory encoding may consistently occur near the *trough* of a theta wave. In neurodegenerative conditions, this structure may degrade, leading to disorganized communication between brain regions.

PAC can be interpreted as a biological form of amplitude modulation (AM): the low-frequency phase acts as the carrier, and the high-frequency amplitude envelope is the modulated signal.

By studying how gamma amplitude fluctuates in alignment with theta phase, we aim to uncover differences in brain coordination patterns across olfactory stimuli (chocolate, rose) and subject groups (HC, MCI, AD). Two complementary methods—MVL and MI—are introduced for this analysis.

1.3 Phase and Amplitude Extraction: Wavelet vs. Hilbert Transform

To compute PAC, it is first necessary to extract the instantaneous *phase* of a low-frequency rhythm (such as theta) and the *amplitude envelope* of a higher-frequency rhythm (such as gamma). Two standard approaches for this are the Hilbert transform and wavelet decomposition, each with distinct strengths.

The **Hilbert transform** requires first applying a narrow bandpass filter to isolate the desired frequency component. The analytic signal is then derived, from which the instantaneous phase and amplitude can be directly calculated. This method provides high temporal precision and is computationally efficient, making it well-suited for clean, stationary frequency bands.

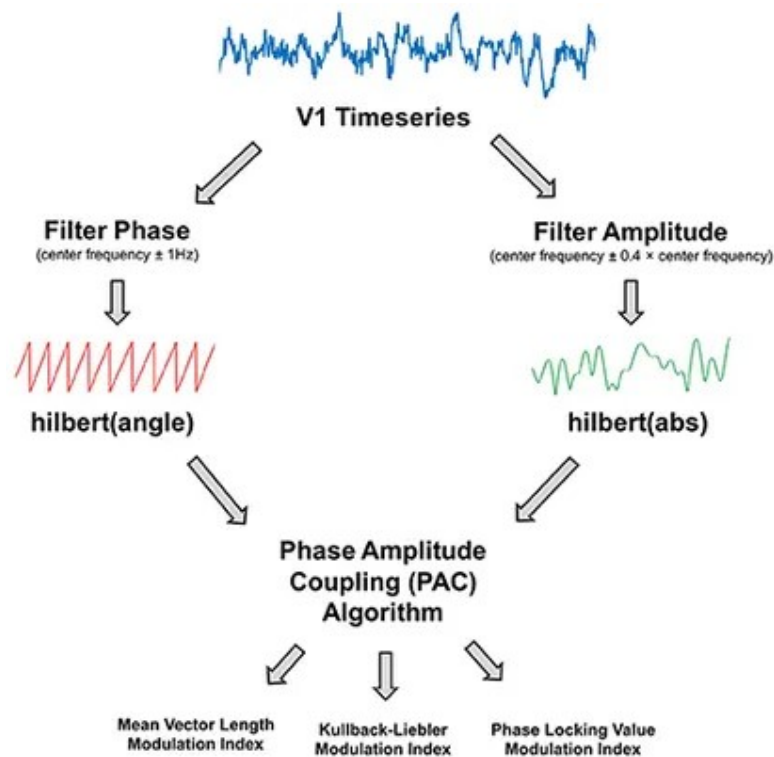


Figure 1: Illustration of the PAC analysis procedure using Hilbert Transform

In contrast, the **wavelet transform**, particularly using complex Morlet wavelets, offers a more flexible and robust time-frequency decomposition. It allows for smooth, continuous tracking of oscillatory activity with better frequency resolution, especially important for non-stationary signals like EEG. Wavelets can handle transient events and overlapping frequency content more gracefully than fixed-window filters.

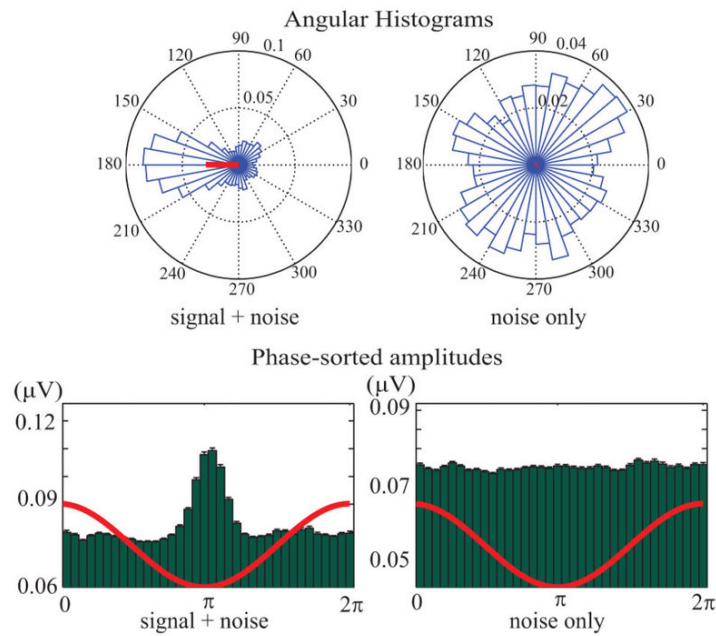


Figure 2: Wavelet Transform Histogram

1.4 PAC via Mean Vector Length (MVL)

Why Use MVL?

The Mean Vector Length (MVL) is a widely used and conceptually straightforward method to quantify PAC. It captures how consistently the amplitude of a high-frequency signal aligns with a specific phase of a slower rhythm. This method is ideal for time-resolved analysis of PAC and allows for direct comparison across trials, stimuli, and groups.

Procedure and Output

To begin, the EEG signal was filtered to isolate the theta band (4–8 Hz) for phase and the gamma band (30–50 Hz) for amplitude. Subsequently, the analytic signal was computed using both wavelet and Hilbert transforms :

Computing the analytic signal using the Hilbert Transform

1.4.1 part of MATLAB CODE:

```
for t = 1:length(trial_i)
    trial = trial_i(t);
    for ch = 1:n_channels
        signal = double(squeeze(EEG.data(ch, :, trial)));
        % filter
        theta_signal = filtfilt(b_theta, a_theta, signal);
        gamma_signal = filtfilt(b_gamma, a_gamma, signal);
        theta_phase = angle(hilbert(theta_signal));
        gamma_amp = abs(hilbert(gamma_signal));

        for w = 1:n_windows
            win_start = start_samples(w);
            win_end = win_start + w_samples - 1;
            phi = theta_phase(win_start:win_end);
            amp = gamma_amp(win_start:win_end);
            z = amp .* exp(1i * phi);
            mvl(ch, w, t) = abs(mean(z));
        end
    end
end
```

1.4.2 Explanation

$$\phi_\theta(t) = \arg[\mathcal{H}(x_\theta(t))], \quad A_\gamma(t) = |\mathcal{H}(x_\gamma(t))|$$

Constructing the complex vector:

$$z(t) = A_\gamma(t) \cdot e^{j\phi_\theta(t)}$$

Computing the MVL over a window of N samples:

$$\text{MVL} = \left| \frac{1}{N} \sum_{t=1}^N z(t) \right|$$

1.4.3 Visualization of MATLAB Output

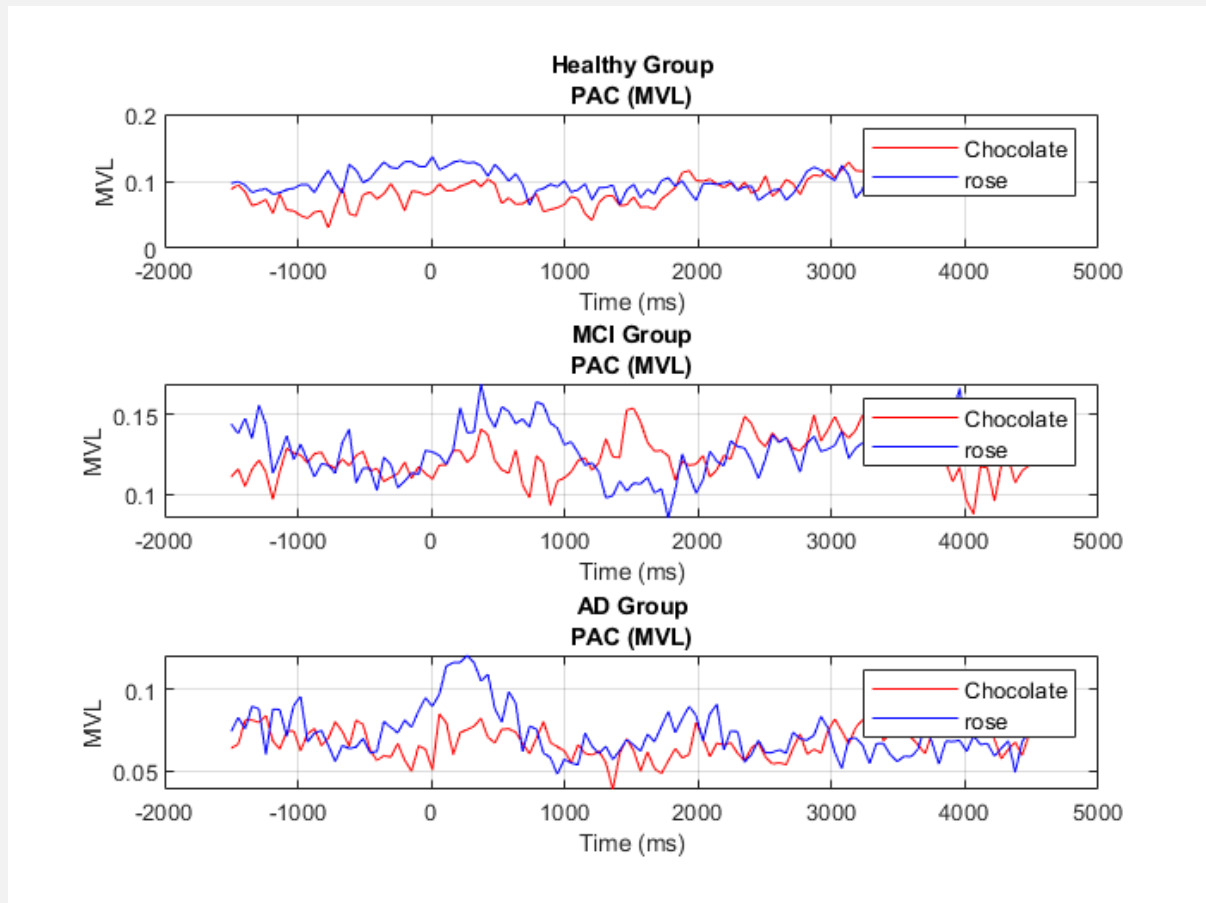


Figure 3: time-resolved MVL curve per subject per condition(channel Cz)

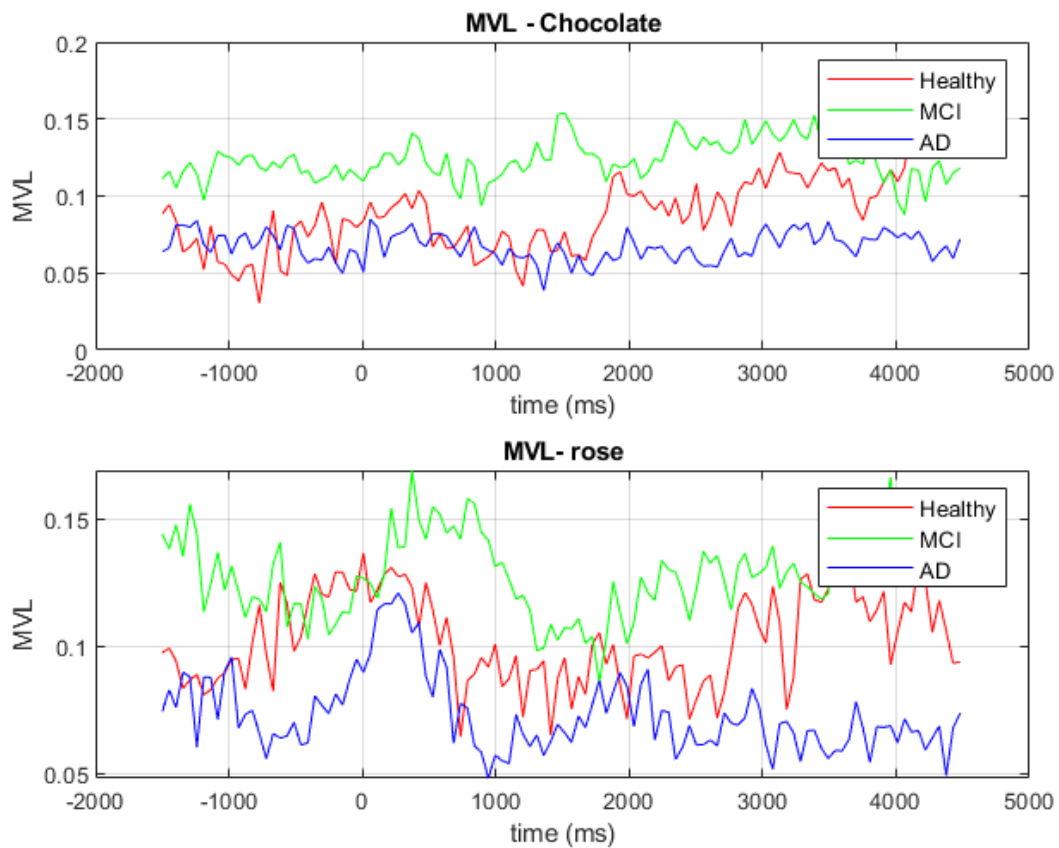


Figure 4: group-level PAC curves for each stimulus (chocolate, rose)(channel Cz)

The analysis above was performed using the Hilbert Transform. In the following section, the same analysis is carried out using the Wavelet Transform to allow for a comparison of their outcomes.

Computing the analytic signal using the Wavelet Transform

1.4.4 part of MATLAB CODE:

```

for t = 1:length(trial_i)
    trial = trial_i(t);
    for ch = 1:n_channels
        signal = double(squeeze(EEG.data(ch, :, trial)));

        %theta_fc=central_frequency=6
        %gamme_fc=central_frequency=40
        %srate=250Hz

        theta_wavelet = morlet_wavelet(theta_fc, 1.8);
        gamma_wavelet = morlet_wavelet(gamma_fc, 1.8);
        theta_conv = conv(signal, theta_wavelet, 'same');
        gamma_conv = conv(signal, gamma_wavelet, 'same');
        theta_phase = angle(theta_conv);
        gamma_amp = abs(gamma_conv);

        for w = 1:n_windows
            win_start = start_samples(w);
            win_end = win_start + w_samples - 1;
            phi = theta_phase(win_start:win_end);
            amp = gamma_amp(win_start:win_end);
            z = amp .* exp(1i * phi);
            mvl(ch, w, t) = abs(mean(z));
        end
    end
end

% Morlet Wavelet Function
function w = morlet_wavelet(fc, sigma)
    t = -4:1/250:4;
    w = (exp(2*1i*pi*fc*t) .* exp(-t.^2 /sigma)) / sqrt(pi*sigma);
end

```

1.4.5 Explanation

The function `morlet_wavelet` takes two arguments as input:

- The central frequency `fc`, which determines the frequency at which the wavelet oscillates
- The time-decay parameter `sigma`, which determines the wavelet's width in time

The output of `morlet_wavelet` is a sine wave oscillating at the center frequency f_c , modulated by a decaying exponential which forms a gaussian envelope, resulting in a complex Morlet Wavelet.

$$w(t) = \frac{1}{\sqrt{\pi\sigma}} \cdot e^{2\pi i f_c t} \cdot e^{-\frac{t^2}{\sigma}}$$

By convolving the Morlet wavelet kernel with the original signal, the following results were obtained:

1.4.6 Visualization of MATLAB Output

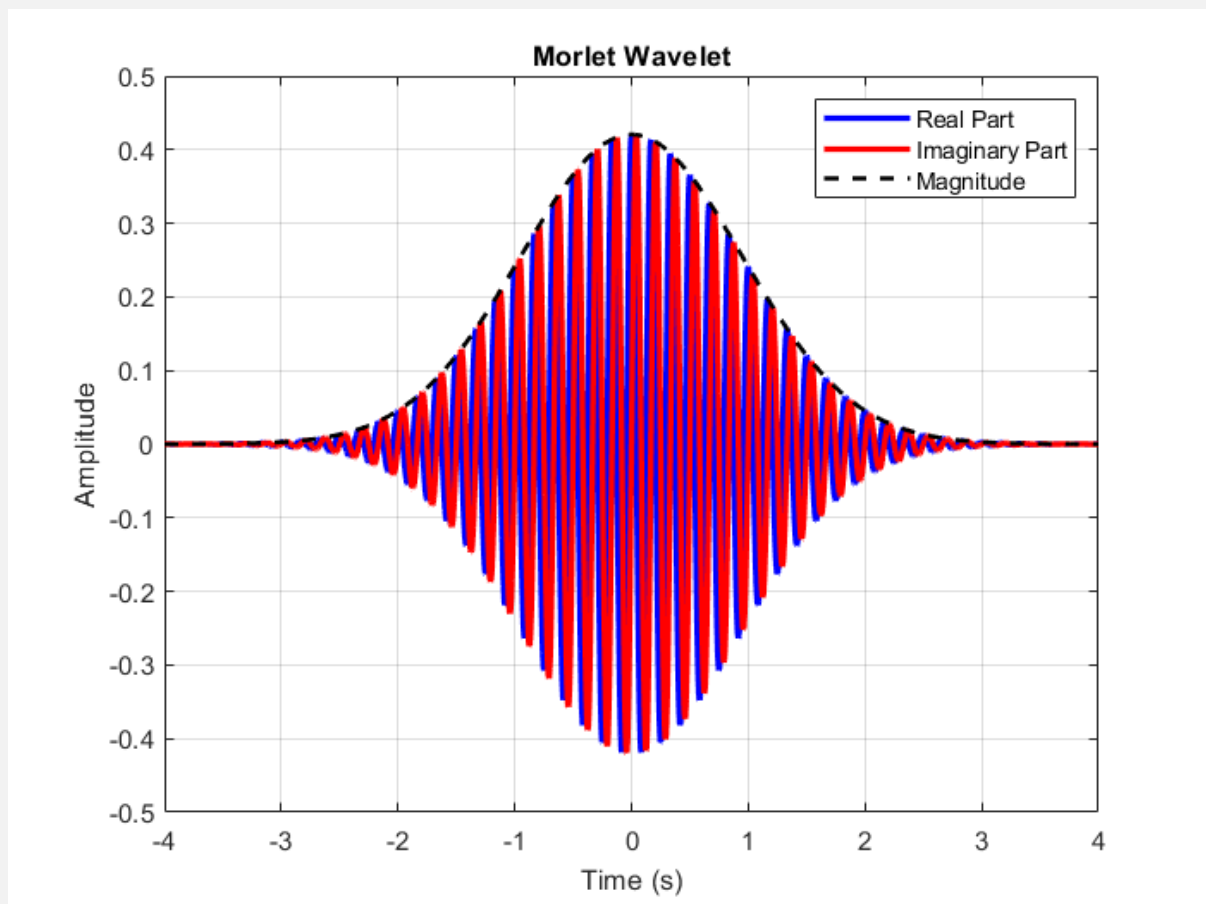


Figure 5: complex Morlet-wavelet

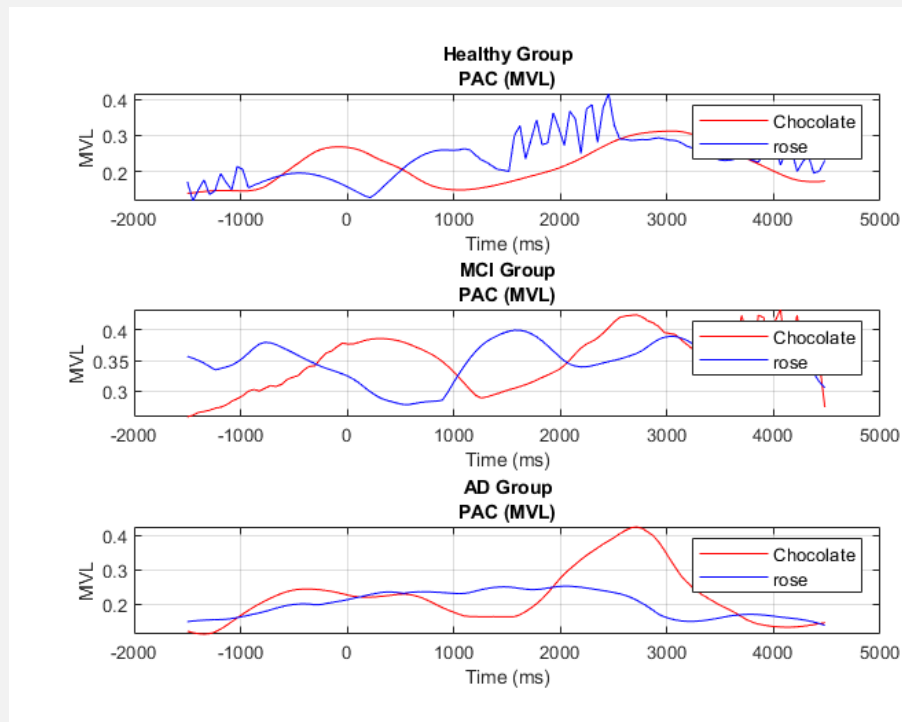


Figure 6: time-resolved MVL curve per subject per condition(channel Cz)

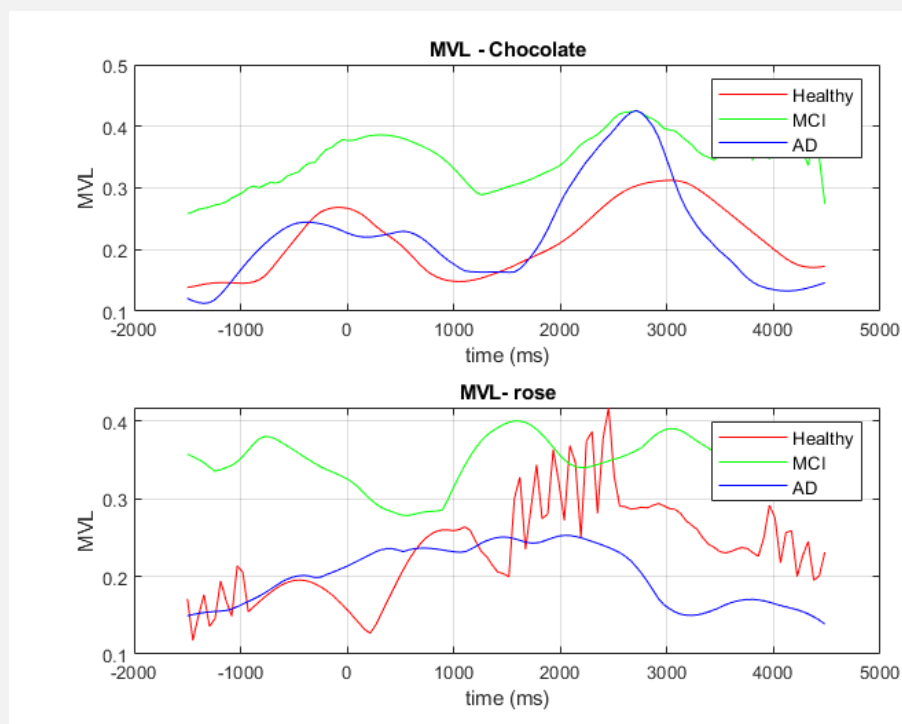


Figure 7: group-level PAC curves for each stimulus (chocolate, rose)(channel Cz)

As the results show, the PAC curves obtained using the Morlet wavelet are smoother and more continuous than those produced by the Hilbert Transform. This demonstrates that the Wavelet Transform handles transient events and non-stationary signals such as EEG more gracefully than fixed-window filters. However, the Hilbert transform results differentiate the subjects' PAC curves more clearly, with the AD group curve being lower than those of the other two groups.

Also, the average PAC values obtained using the wavelet transform appear to be a little higher than those obtained using the Hilbert transform.

Interpretation

- Is there a clear increase in PAC during stimulus delivery?

Looking at the time-resolved MVL curves obtained using the Hilbert Transform (Figure 4), we observe that at time $t = 0$, the PAC for the Healthy and MCI groups show a clear increase followed by a reduction, compared to the AD group. The AD group oscillates around a relatively constant value (baseline value), without exhibiting a distinct peak, and its average PAC value is lower than that of the other two groups. The expected result was an increased PAC (higher MVL) for the Healthy group; however, the results obtained, show a stronger time-locked PAC for the MCI group. This still supports the prediction that the AD group tends to have a low MVL value. The deviation could be due to the preprocessing stage or the data gathering conditions. For the chocolate odor, the average MVL value is higher compared to the rose odor, and the PAC curves show a steadier trend.

- Are coupling dynamics consistent across trials and subjects?

The results were obtained after averaging over trials for each subject and each odor. Such consistency is not clearly evident, as the curves differ between subjects. For example, the AD group shows a relatively steadier curve compared to the other two groups. The curves also vary between odors: all groups show stronger PAC for chocolate, with the MCI group performing better in this condition. For the rose odor, the Healthy group exhibits a greater increase in PAC.

- Does the PAC pattern distinguish between cognitive states?

Based on the obtained results, the MCI group shows the highest MVL value, which is likely an error caused by the preprocessing or data gathering stages. The Healthy group's PAC value is relatively close to that of the MCI group, while the AD group has the lowest MVL value. In conclusion, the AD group can be easily distinguished from the other two; however, the Healthy and MCI groups exhibit very similar patterns, with the MCI group being slightly higher, making them more difficult to classify.

1.5 Bonus Part: PAC via Modulation Index (MI)

Why Use MI?

While MVL summarizes overall coupling strength, it does not reveal how gamma amplitude is distributed across the full phase cycle of a slower oscillation. The Modulation Index (MI), introduced by Tort et al., captures this structural aspect by measuring the deviation of the phase-amplitude distribution from uniformity. This enables a more detailed characterization of the coupling and supports visualization through polar (circular) histograms. MI is particularly useful when interpreting whether certain frequencies preferentially interact at specific phases and how this differs across brain regions or cognitive states.

Procedure and Output

To compute MI, The EEG data is filtered into the theta band (4–8 Hz) and gamma band (30–50 Hz), then instantaneous phase and amplitude is extracted using the Hilbert transform. The reason for using the Hilbert Transform instead of the Wavelet Transform is its simpler arithmetic and the availability of a built-in function for Hilbert, whereas the Wavelet Transform must be defined from scratch.

Next, the phase range $(-\pi, \pi]$ is divided into n equal-width bins ($n = 18$). For each bin θ_i , the average gamma amplitude A_i corresponding to the theta phases that fall within that bin is calculated. The resulting amplitudes are then normalized to obtain a probability distribution:

$$P(\theta_i) = \frac{A_i}{\sum_{j=1}^n A_j}$$

The Modulation Index is then computed as the Kullback-Leibler divergence between this empirical distribution and a uniform one:

$$MI = \frac{D_{KL}(P(\theta) \parallel U(\theta))}{\log(n)}$$

$$D_{KL}(P \parallel U) = \sum_i P(i) \cdot \log \left(\frac{P(i)}{U(i)} \right)$$

where D_{KL} represents the Kullback-Leibler divergence and $U(\theta)$ is the uniform distribution.

PAC via Modulation Index(MI)

1.5.1 part of MATLAB CODE:

```

n_channels = EEG.nbchan;
n_trials = length(EEG.trials);
trial_i = EEG.trials;
trials_MI = zeros(n_trials , n_channels);
all_bin_amp = zeros(n_trials , n_bins , n_channels);
parts = linspace(-pi , pi , n_bins+1);

for ch = 1:n_channels
    for t = 1:length(trial_i)
        trial = trial_i(t);
        signal = double(squeeze(EEG.data(ch, :, trial)));
        % Filtering
        theta_signal = filtfilt(b_theta , a_theta , signal);
        gamma_signal = filtfilt(b_gamma , a_gamma , signal);
        theta_phase = angle(hilbert(theta_signal));
        gamma_amp = abs(hilbert(gamma_signal));

        bin_amp = zeros(1 , n_bins);
        for n = 1:n_bins
            i = theta_phase >= parts(n) & theta_phase < parts(n+1);
            bin_amp(n) = mean(gamma_amp(i));
        end

        p = bin_amp / sum(bin_amp);
        uniform_distr = ones(1 , n_bins) / n_bins;
        D_KL = sum(p .* log(p ./ uniform_distr));
        MI = D_KL / log(n_bins);

        trials_MI(t , ch) = MI;
        all_bin_amp(t , :, ch) = bin_amp;
    end
end

avg_bin_amp = squeeze(mean(all_bin_amp , 1));
mean_MI = mean(trials_MI , 1);
theta_centers = (parts(1:end-1) + parts(2:end)) / 2;

mi_results.(groups{g}).(stimuli{s}).MI_per_trial = trials_MI;
mi_results.(groups{g}).(stimuli{s}).mean_MI = mean_MI;
mi_results.(groups{g}).(stimuli{s}).theta_centers = theta_centers;
mi_results.(groups{g}).(stimuli{s}).avg_bin_amp = avg_bin_amp;

```

Visualization of MATLAB Output

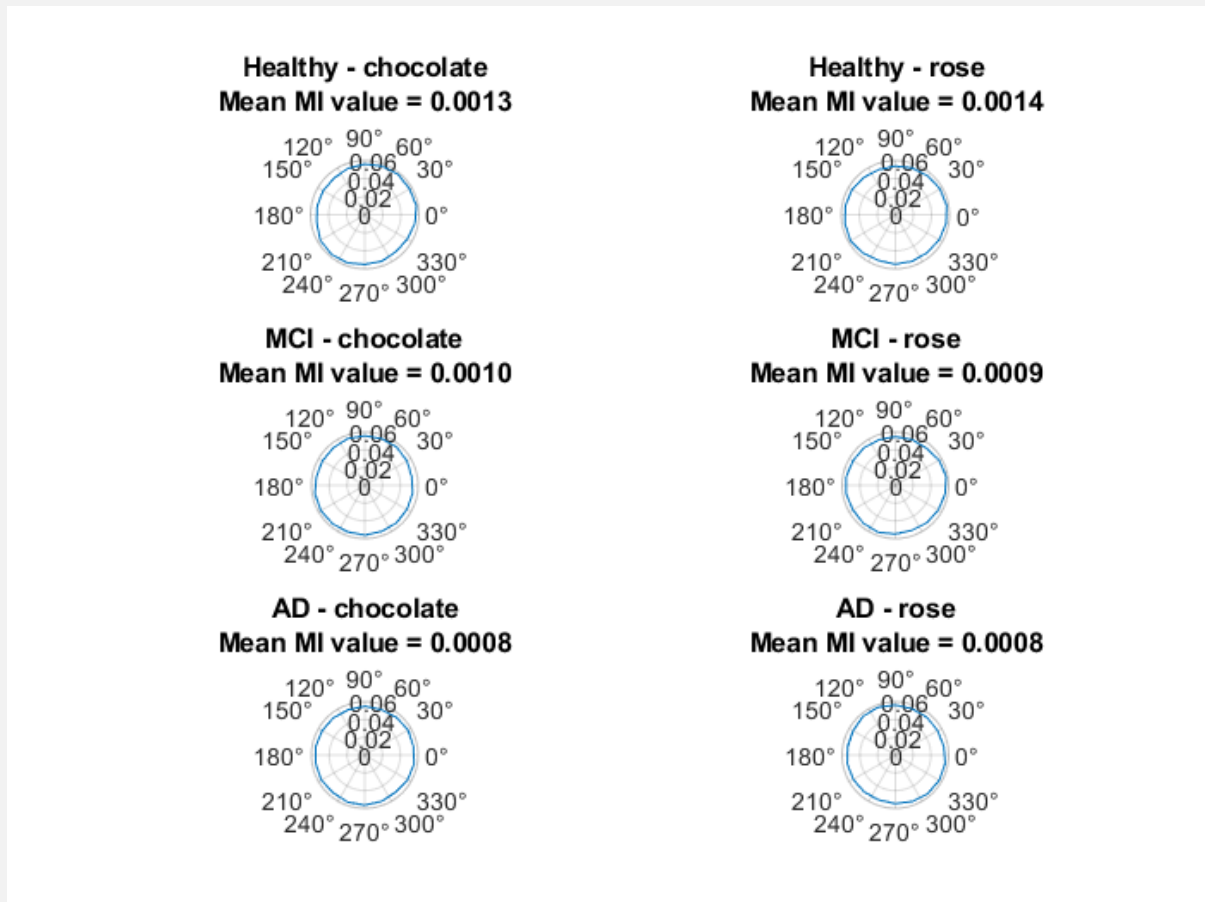


Figure 8: polar histogram of amplitude vs. phase(average over all channels)

Interpretation of MI Output

Based on the results, the average MI value across all channels is highest for the Healthy group, while the AD group shows the lowest mean MI value, which aligns with expectations.

The Healthy group was expected to show a clear peak, indicating structured coupling; however, the plot appears relatively flat, with no distinct peak observed. In contrast, the MCI and AD groups were expected to show flatter plots, reflecting degraded coupling — which is consistent with the observed results. The observed error may have resulted from the data collection process or from the preprocessing settings used.

Electrode Selection and Anatomical Considerations

For additional insight, examining PAC across electrode sites aligned with known processing pathways was considered. Using frontal (e.g., **Fz**) and parietal (e.g., **Pz**) electrodes allows investigation of PAC along the rostro-caudal axis. Coupling where theta phase is from Fz and gamma amplitude from Pz may reflect a *feedforward* interaction, while the reverse could suggest *feedback* communication.

Theta Phase (Fz) – Gamma Amplitude (Pz) Coupling

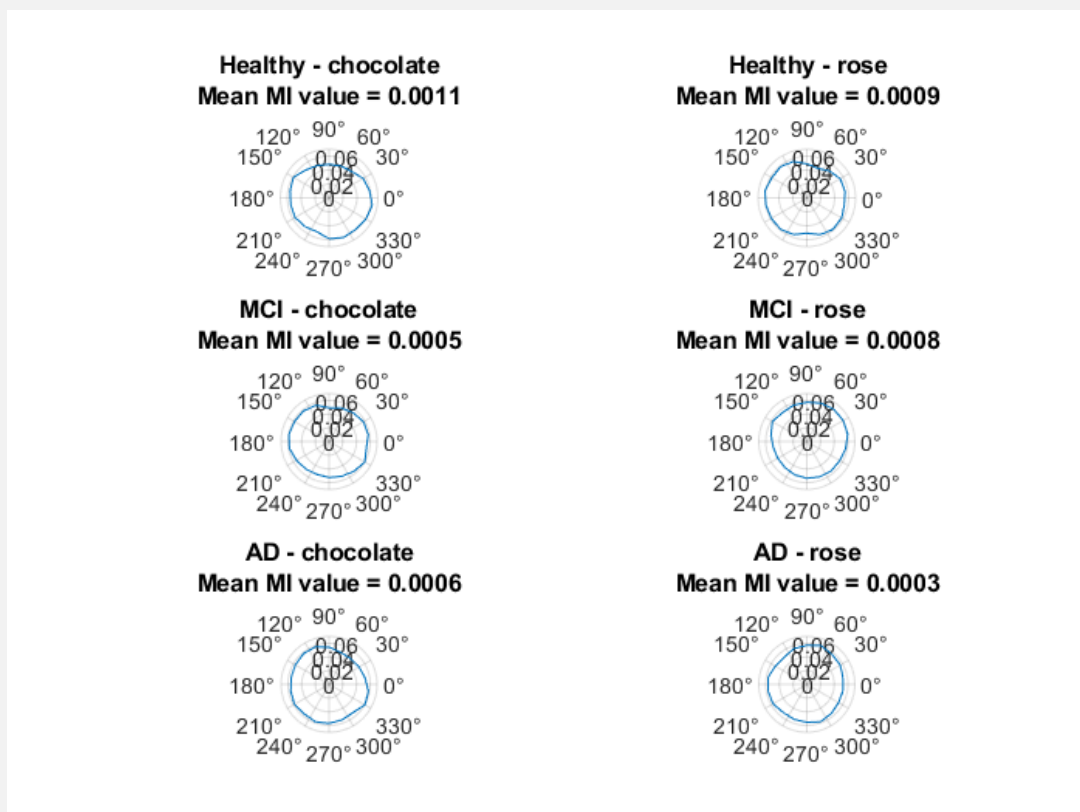


Figure 9: polar histogram of amplitude vs. phase (*Theta Phase (Fz) – Gamma Amplitude (Pz)*)

Feedforward interaction refers to the flow of information from a lower (earlier) processing region toward a later-processing region. In this examination, the theta phase is taken from Fz (frontal cortex) and the gamma amplitude from Pz (parietal cortex).

From the results, the mean MI values for the rose stimulus decrease progressively from the Healthy group to the AD group. For the chocolate odor, the Healthy group shows the highest MI, whereas the MCI group has the lowest MI instead of the AD group.

Overall, we can conclude that the feedforward interaction is strongest in the Healthy group compared to the other two groups.

Theta Phase (Pz) – Gamma Amplitude (Fz) Coupling

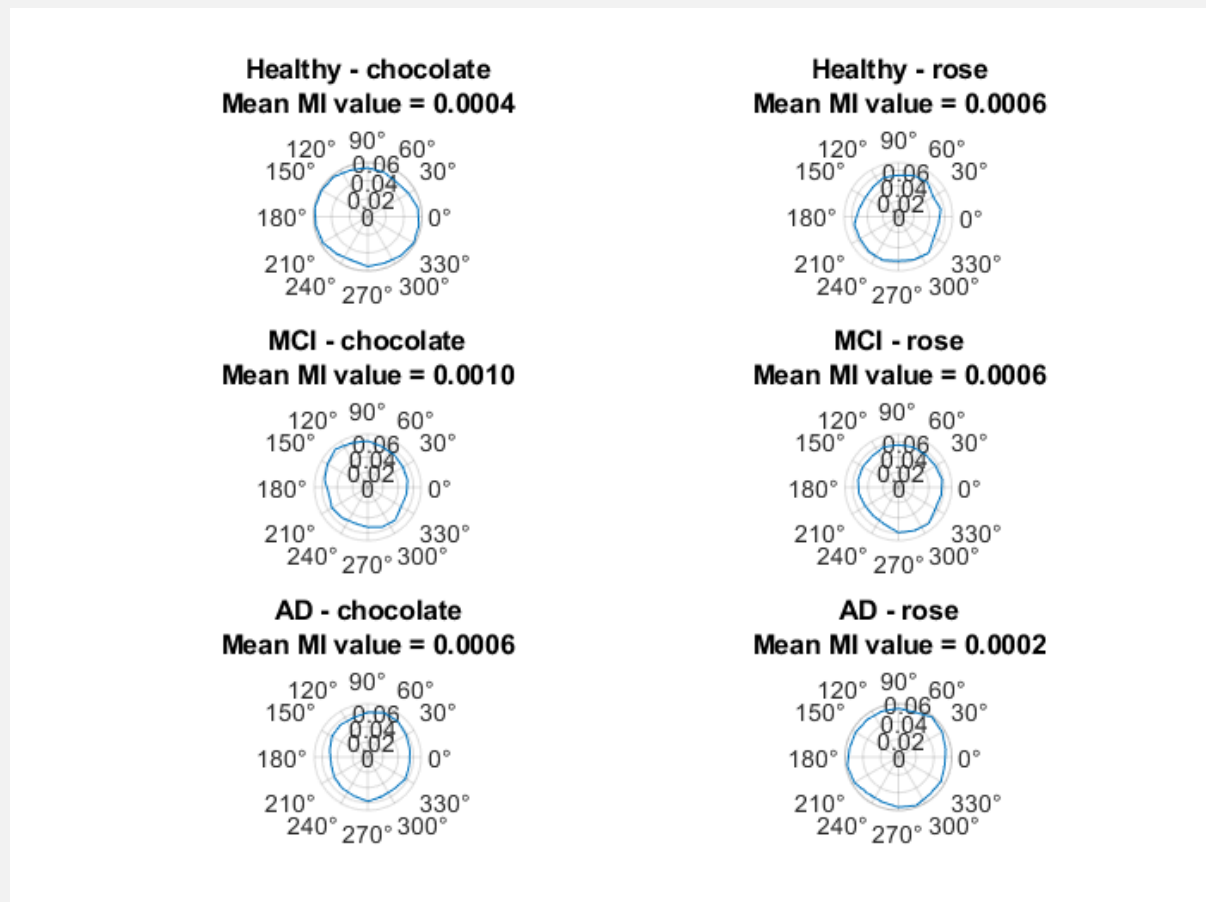


Figure 10: polar histogram of amplitude vs. phase (*Theta Phase (Pz) – Gamma Amplitude (Fz)*)

Feedback interaction refers to the flow of information from a later-processing region toward an earlier-processing region. In this examination, the theta phase is taken from Pz (parietal cortex) and the gamma amplitude from Fz (frontal cortex).

From the results, for the rose stimulus, the mean MI values are similar for the Healthy and MCI groups, while the AD group has the lowest value. For the chocolate odor, the MCI group shows the highest MI, whereas the Healthy group has the lowest, instead of the AD group.

Overall, we can conclude that the Healthy group has stronger feedforward interaction than feedback, although the difference is small. In the AD group, both interactions are almost equal, and for the MCI group, feedback is stronger for the chocolate odor, while feedforward is stronger for the rose odor.

Local PAC at single electrodes

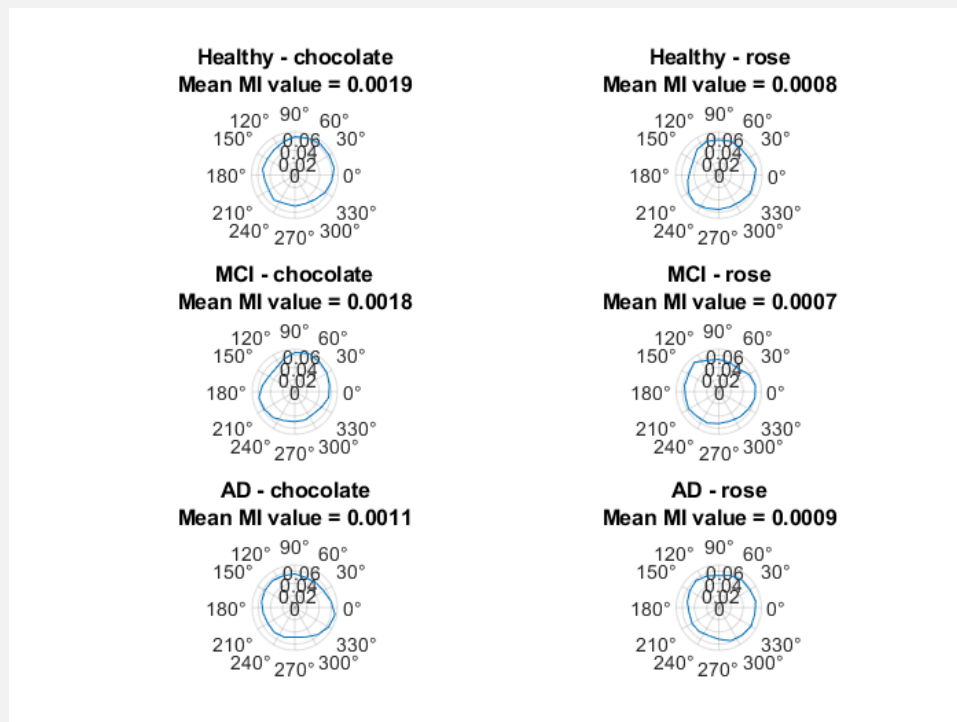


Figure 11: polar histogram of amplitude vs. phase(*channel Fz*)

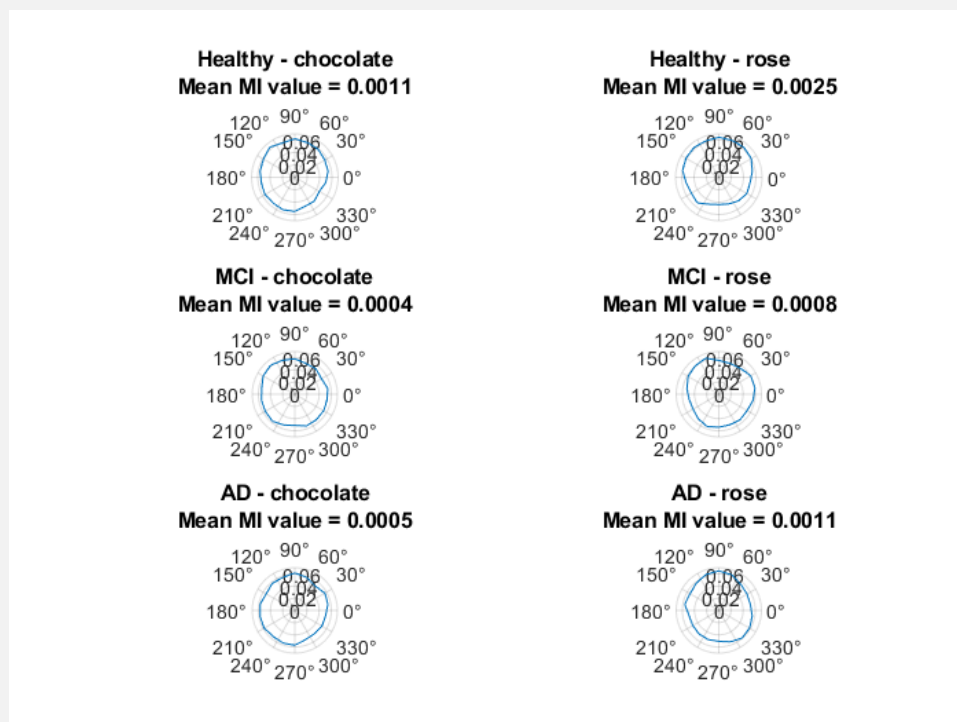


Figure 12: polar histogram of amplitude vs. phase(*channel Pz*)

Local PAC at single electrodes

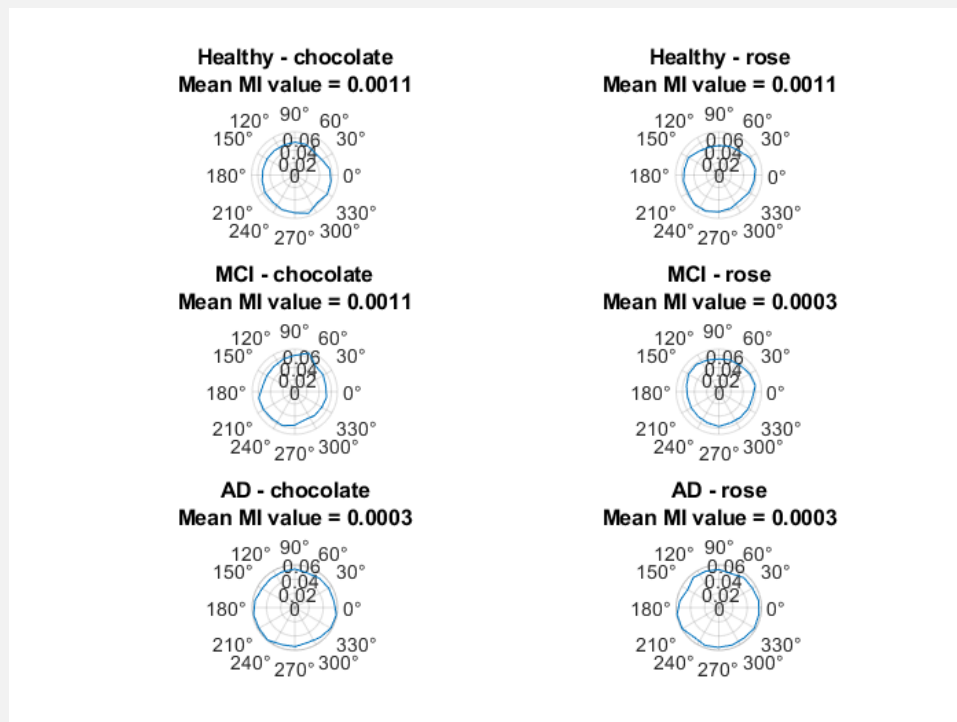


Figure 13: polar histogram of amplitude vs. phase(*channel FP1*)

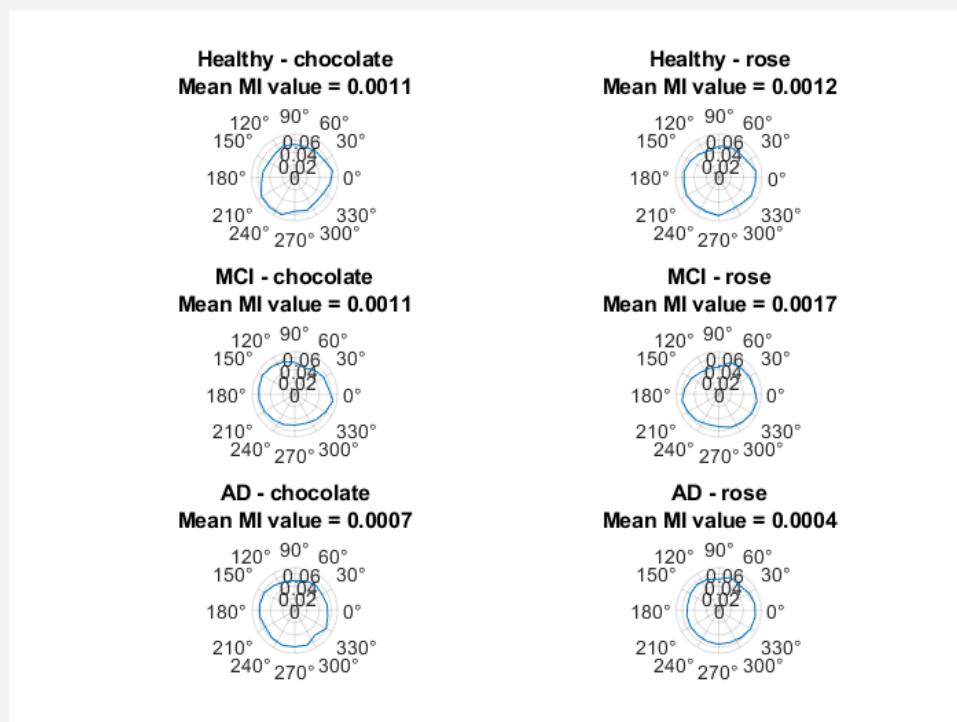


Figure 14: polar histogram of amplitude vs. phase(*channel FP2*)

Interpretation

- MI-based analyses complement MVL by clarifying the structure and specificity of PAC. In addition to evaluating MI values across odors and subject groups, consider how phase preferences are organized across the scalp. Are gamma bursts aligned with particular theta phases? Do polar plots reveal consistent coupling in healthy subjects but disrupted or less focal patterns in MCI or AD?

As observed from the figures above, the polar histograms do not show significant coupling and appear relatively flat. However, when comparing across groups based on the outputs from local electrodes, the Healthy group tends to display a more distorted pattern and also shows the highest average MI values in most cases. The AD group, in contrast, appears flatter, with no distinguishable peak, and looks almost circular. The MCI group behaves similarly to the Healthy group in some cases, showing small couplings. The slight peaks are not necessarily aligned with specific phases and occur at different angles each time, for example, in Figure 13, the coupling for the Healthy group with the chocolate stimulus occurs at phase zero.

- You may also explore how MI-based observations correspond to MVL time courses—do strong MVL peaks coincide with sharply defined phase distributions? This multi-angle approach can offer deeper insights into how rhythmic coordination is altered in neurodegeneration.

MVL and MI are both consistent measures of PAC, and therefore their results should largely agree. If noise does not have a significant impact, the peaks of the MVL curves are expected to coincide with phase–amplitude couplings in the polar histograms.

In our case, the MVL curves fluctuate around a baseline with slight peaks, and the polar histograms show a similar pattern. Moreover, in both MVL curves and polar histograms, the Healthy and MCI groups have closely similar trends, performing better than the AD group. The Healthy and MCI groups also exhibit higher average amplitude values and more distinguishable peaks compared to AD.

1.6 Conclusion

This phase of the project explores how oscillatory components of brain activity interact during olfactory processing through the lens of Phase-Amplitude Coupling. By first establishing a dynamic baseline using MVL, and optionally expanding the analysis with MI, a multifaceted perspective on the organization of brain rhythms emerges.

MVL serves as a robust, time-resolved measure of PAC strength, offering immediate insight into whether coupling increases with stimulus onset. MI complements this view by providing a detailed structural profile of how amplitude is distributed across phase, and supports intuitive interpretation through visual plots.

Together, these methods allow for the exploration of coordinated brain dynamics with greater nuance than power analysis alone. They help reveal whether disruptions in PAC might serve as early electrophysiological indicators of cognitive impairment, and demonstrate the value of signal processing tools in neuroscience research.

References

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